# 4-yearly maintenance

ENERCON
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## Master data

| | | | |
|---|---|---|---|
| **Serial number** | 822882 | **Plant** | S-03855-003 |
| **Type** | E-82 E2 | **Location** | Ger |
| **Control type** | CS82a | | |

## Order

| | |
|---|---|
| **Order type** | 4-yearly maintenance |
| **Order number** | 95844445 |
| **Confirmation date** | Jun 14, 2019 |
| **Report number** | 940000863100 |
| **Document number** | TD-esc-08-xxx-xxx-15-055 Rev007 |

# Inspection report

## Summary

| | | | | | |
|---|---|---|---|---|---|
| **Without findings** | 261 | **With findings** | 11 | **Total** | 272 |

## Details

| No. | Work step | Result |
|---|---|---|
| | **Electrics - Tower base** | |
| 1 | Check tower base warning systems | 3 Not done - Not necessary |
| 2 | Updating software | 2 Done - Defects detected |
| 3 | Checking the WEC parameters and operating parameters | 1 Done - No defects |
| 4 | Checking the rechargeable battery of the ENERCON SCADA Server UPS | 1 Done - No defects |
| 5 | Check access control | 3 Not done - Not necessary |
| 6 | Checking the shadow shutdown light sensors | 3 Not done - Not necessary |
| 7 | Measuring earth resistance | 1 Done - No defects |
| 8 | Measuring rotor blade de-icing currents | 3 Not done - Not necessary |
| | **Electrics - External transformer station** | |
| 9 | Checking sockets in external transformer station | 3 Not done - Not necessary |
| 10 | Measuring insulation resistance on sockets in external transformer station | 3 Not done - Not necessary |
| 11 | Measuring the earth resistance - external transformer station | 3 Not done - Not necessary |
| 12 | Checking capacitive voltage taps in external transformer station | 3 Not done - Not necessary |
| 13 | Checking transformer room in external transformer station | 3 Not done - Not necessary |
| 14 | Visually checking transformer in external transformer station | 3 Not done - Not necessary |
| 15 | Taking an oil sample in external transformer station | 3 Not done - Not necessary |

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| No. | Work step | Result |
|---|---|---|
| 16 | Checking medium-voltage switchgear in external transformer station | 3 Not done - Not necessary |
| 17 | Checking medium-voltage switchgear (every four years) in external transformer station | 3 Not done - Not necessary |
| 18 | Checking low-voltage distribution system in external transformer sta- tion | 3 Not done - Not necessary |
| 19 | Checking terminal lugs of low-voltage distribution system - external transformer station | 3 Not done - Not necessary |
| 20 | Checking station basement in external transformer station | 3 Not done - Not necessary |
| | **Electrics - WEC medium voltage area** | |
| 21 | Checking the capacitive voltage taps | 1 Done - No defects |
| 22 | Checking the transformer room | 1 Done - No defects |
| 23 | Carrying out a visual inspection of the transformer | 1 Done - No defects |
| 24 | Taking and oil sample | 3 Not done - Not necessary |
| 25 | Checking medium-voltage switchgear | 1 Done - No defects |
| 26 | Checking the medium-voltage switchgear every 4 years | 1 Done - No defects |
| 27 | Checking the low-voltage distribution system | 1 Done - No defects |
| 28 | Checking LV distribution system fans | 3 Not done - Not necessary |
| 29 | Checking low-voltage distribution system terminal lugs | 3 Not done - Not necessary |
| 30 | Checking the terminal lugs for the secondary systems cabinet | 3 Not done - Not necessary |
| 31 | Checking control cabinet terminal lugs | 1 Done - No defects |
| 32 | Checking UPS terminal lugs | 1 Done - No defects |
| 33 | Checking NH1 fuse switch disconnector terminal lugs | 3 Not done - Not necessary |
| 34 | Checking the foundation basement | 1 Done - No defects |
| 35 | Checking AC busbar system V3 | 3 Not done - Not necessary |
| | **Electrics - Tower base** | |
| 36 | Measuring continuity of protective conductor system | 1 Done - No defects |
| 37 | Measuring insulation resistance | 1 Done - No defects |
| 38 | Checking electrical cabinets | 1 Done - No defects |
| 39 | Checking the electrical cabinets overvoltage protection | 1 Done - No defects |
| 40 | Checking the SCADA interface backup battery | 1 Done - No defects |
| | **Electrics - External transformer station** | |
| 41 | Measuring continuity on protective conductor system in external trans- former station | 3 Not done - Not necessary |
| 42 | Performing electrical check on RCD in external transformer station | 3 Not done - Not necessary |
| | **Electrics - WEC medium voltage area** | |
| 43 | Checking the ENERCON transformer protection system | 1 Done - No defects |
| | **Electrics - Tower base** | |
| 44 | Checking the tower cooling system | 1 Done - No defects |

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|---|---|---|
| 45 | Checking the UPS | 1 Done - No defects |
| 46 | Checking DC charger unit | 3 Not done - Not necessary |
| 47 | Checking UPS heating | 1 Done - No defects |
| 48 | Checking the heating system for the power cabinets and control cab- inet | 1 Done - No defects |
| 49 | Checking the power cabinet fans | 1 Done - No defects |
| 50 | Testing emergency lights | 1 Done - No defects |
| 51 | Performing an electrical inspection on the RCD | 1 Done - No defects |
| 52 | Performing an electrical inspection on an upstream RCD | 3 Not done - Not necessary |
| 53 | Measuring short circuit current | 1 Done - No defects |
| 54 | Checking the auxiliary contacts | 3 Not done - Not necessary |
| 55 | Checking the LV earth fault detection system current sensor | 3 Not done - Not necessary |
| 56 | Checking the service hoist emergency lighting system | 1 Done - No defects |
| 57 | Performing an electrical inspection on the service hoist RCD | 1 Done - No defects |
| 58 | Checking sockets | 1 Done - No defects |
| 59 | Measuring insulation resistance of sockets | 1 Done - No defects |
| | **Electrics - Tower** | |
| 60 | Checking sockets | 2 Done - Defects detected |
| 61 | Measuring insulation resistance of sockets | 1 Done - No defects |
| 62 | Performing an electrical inspection on the RCD | 1 Done - No defects |
| | **Electrics - Machine house** | |
| 63 | Nacelle warning systems | 3 Not done - Not necessary |
| 64 | Checking the horns | 1 Done - No defects |
| 65 | Checking the telephone connection | 1 Done - No defects |
| 66 | Nacelle orientation to north and cable twisting | 1 Done - No defects |
| 67 | Checking the rotor lock | 1 Done - No defects |
| 68 | Checking rotor blade de-icing system | 3 Not done - Not necessary |
| 69 | Performing an electrical functional check on the winch | 1 Done - No defects |
| 70 | Checking sockets | 1 Done - No defects |
| 71 | Measuring insulation resistance of sockets | 1 Done - No defects |
| 72 | Testing emergency lights | 1 Done - No defects |
| 73 | Performing an electrical inspection on the RCD | 1 Done - No defects |
| 74 | Checking the yaw drives | 1 Done - No defects |
| 75 | Checking the smoke detectors | 1 Done - No defects |
| 76 | Checking the nacelle noise sensor | 3 Not done - Not necessary |
| 77 | Checking the nacelle control cabinet fan | 1 Done - No defects |

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| 78 | Checking the nacelle control cabinet heating | 1 Done - No defects |
| 79 | Checking the excitation controller fan | 1 Done - No defects |
| 80 | Checking heater in excitation controller box | 1 Done - No defects |
| 81 | Checking the yaw inverter cabinet heating | 1 Done - No defects |
| 82 | Checking the generator filter cabinet fan | 1 Done - No defects |
| 83 | Checking the generator filter cabinet terminal lugs | 2 Done - Defects detected |
| 84 | Checking the rectifier fan | 1 Done - No defects |
| 85 | Checking the rectifier heating | 1 Done - No defects |
| 86 | Checking the auxiliary contacts | 1 Done - No defects |
| 87 | Checking the cable twist limit switch | 1 Done - No defects |
| 88 | Checking cam limit switch | 3 Not done - Not necessary |
| 89 | Checking electrical cabinets | 2 Done - Defects detected |
| 90 | Checking the electrical cabinets overvoltage protection | 1 Done - No defects |
| 91 | Measuring short circuit current | 1 Done - No defects |
| 92 | Measuring continuity of protective conductor system | 1 Done - No defects |
| 93 | Testing torque monitoring system | 1 Done - No defects |
| 94 | Polarisation index test | 1 Done - No defects |
| | **Electrics - Rotor head** | |
| 95 | Testing lights | 1 Done - No defects |
| 96 | Checking electrical cabinets | 1 Done - No defects |
| 97 | Checking the electrical cabinets overvoltage protection | 1 Done - No defects |
| 98 | Measuring short circuit current | 1 Done - No defects |
| 99 | Measuring continuity of protective conductor system | 1 Done - No defects |
| 100 | Checking sockets | 1 Done - No defects |
| 101 | Measuring insulation resistance of sockets | 1 Done - No defects |
| 102 | Performing an electrical inspection on the RCD | 1 Done - No defects |
| 103 | Checking the Cobham slip ring unit and Label slip ring unit | 1 Done - No defects |
| 104 | Checking the ENERCON slip ring unit | 3 Not done - Not necessary |
| 105 | Checking the torque support and drive mechanism | 1 Done - No defects |
| 106 | Checking the generator | 1 Done - No defects |
| 107 | Checking rotor monitoring system | 1 Done - No defects |
| 108 | Checking rotor head noise sensor | 1 Done - No defects |
| 109 | Checking the air gap switch | 1 Done - No defects |
| 110 | Checking rotor blade A’s limit switch | 1 Done - No defects |
| 111 | Checking rotor blade B’s limit switch | 1 Done - No defects |

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| 112 | Checking rotor blade C's limit switch | 1 Done - No defects |
| 113 | Checking pitch motor fans of rotor blade A | 3 Not done - Not necessary |
| 114 | Checking pitch motor fans of rotor blade B | 3 Not done - Not necessary |
| 115 | Checking pitch motor fans of rotor blade C | 3 Not done - Not necessary |
| 116 | Checking the rotor head discharge box | 2 Done - Defects detected |
| 117 | Checking discharge box of rotor blade A | 1 Done - No defects |
| 118 | Checking discharge box of rotor blade B | 1 Done - No defects |
| 119 | Checking discharge box of rotor blade C | 1 Done - No defects |
| | **Electrics - Attachment parts outside the** | |
| 120 | Checking the ultrasonic anemometer alignment | 3 Not done - Not necessary |
| 121 | Check precipitation sensor | 3 Not done - Not necessary |
| 122 | Checking the beacon system | 1 Done - No defects |
| | **Electrics - Machine house** | |
| 123 | Checking air cooling system | 1 Done - No defects |
| 124 | Checking the rectifier earth fault monitoring system | 1 Done - No defects |
| 125 | Checking generator residual current monitoring | 1 Done - No defects |
| | **Electrics - Emergency shutdowns** | |
| 126 | Checking the nacelle control cabinet EMERGENCY STOP buttons | 1 Done - No defects |
| 127 | Testing the control cabinet EMERGENCY STOP button | 1 Done - No defects |
| 128 | Testing the tower entrance door EMERGENCY STOP button | 1 Done - No defects |
| 129 | Testing the E-module EMERGENCY STOP button | 3 Not done - Not necessary |
| 130 | Checking the main safety circuit relay K4 emergency shutdown | 1 Done - No defects |
| 131 | Checking the main safety circuit relay K3 emergency shutdown | 1 Done - No defects |
| 132 | Checking the rotor lock emergency shutdown | 1 Done - No defects |
| 133 | Checking the vibration sensor emergency shutdown | 1 Done - No defects |
| | **Electrics - Machine house** | |
| 134 | Measuring rotor blade de-icing currents | 3 Not done - Not necessary |
| | **Mechanical - Tower base** | |
| 135 | Visual inspection of tower base | 1 Done - No defects |
| 136 | Checking foundation | 1 Done - No defects |
| 137 | Checking bolt connections on bolt cage with load ring | 3 Not done - Not necessary |
| 138 | Checking fire extinguisher in tower base | 1 Done - No defects |
| | **Mechanical - Tower** | |
| 139 | Checking the tower anchorage points | 1 Done - No defects |
| 140 | Checking the tower ventilation system | 1 Done - No defects |

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| 141 | Checking the tower bolt connections | 1 Done - No defects |
| 142 | Checking the concrete tower | 1 Done - No defects |
| 143 | Checking the prestressing system | 1 Done - No defects |
| 144 | Checking steel tower | 1 Done - No defects |
| 145 | Checking steel section | 1 Done - No defects |
| 146 | Visual inspection of tower | 1 Done - No defects |
| | **Mechanical - Safety ladder** | |
| 147 | Checking the safety ladder stickers | 1 Done - No defects |
| 148 | Checking start and end of climbing route | 1 Done - No defects |
| 149 | Check C-profile and fall arrest noses | 1 Done - No defects |
| 150 | Checking fastening brackets and clamps | 2 Done - Defects detected |
| 151 | Checking rungs | 1 Done - No defects |
| 152 | Checking gap widths on ladder sections | 1 Done - No defects |
| 153 | Visually inspecting safety ladder | 1 Done - No defects |
| 154 | Checking safety catches | 1 Done - No defects |
| 155 | Checking the lateral rail exits | 1 Done - No defects |
| 156 | Checking the small rest platforms | 1 Done - No defects |
| 157 | Checking the safety ladder fasteners | 1 Done - No defects |
| 158 | Performing a test climb | 1 Done - No defects |
| 159 | Applying “Next Check” inspection label | 1 Done - No defects |
| | **Mechanical - Service hoist** | |
| 160 | Checking that the document pouch is complete | 1 Done - No defects |
| 161 | Checking the safety signs | 1 Done - No defects |
| 162 | Checking the lift cage | 1 Done - No defects |
| 163 | Checking the anchorage points | 1 Done - No defects |
| 164 | Checking the cable guides | 1 Done - No defects |
| 165 | Checking the key-operated switch in the lift cage | 1 Done - No defects |
| 166 | Checking the safety guard and cable pulley in the lift cage | 1 Done - No defects |
| 167 | Checking the lift cage crossbeam | 1 Done - No defects |
| 168 | Checking the door safety switch | 1 Done - No defects |
| 169 | Checking the top limit switch | 1 Done - No defects |
| 170 | Checking the bottom limit switches | 1 Done - No defects |
| 171 | Checking the bottom outside sensor | 3 Not done - Not necessary |
| 172 | Checking the emergency stop button | 1 Done - No defects |
| 173 | Checking the emergency control unit | 1 Done - No defects |

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| 174 | Checking the hoist cable and safety cable at the hoist touchdown platform | 1 Done - No defects |
| 175 | Checking the cable fasteners with landing box | 1 Done - No defects |
| 176 | Checking the cable fasteners without landing box | 3 Not done - Not necessary |
| 177 | Checking the tension of the guide cables | 1 Done - No defects |
| 178 | Checking the lighting in the lift cage | 1 Done - No defects |
| 179 | Checking the indicator lights | 1 Done - No defects |
| 180 | Checking service hoist winch | 1 Done - No defects |
| 181 | Checking the safety catch | 1 Done - No defects |
| 182 | Check automatic operation | 4 Not done - Assembly suspended |
| 183 | Checking the steel cables | 1 Done - No defects |
| 184 | Oiling the wire ropes | 1 Done - No defects |
| 185 | Checking the cable guides at the landing platforms | 1 Done - No defects |
| 186 | Checking the trailing cable and the restraint | 1 Done - No defects |
| 187 | Checking the landing platform switches on the platforms | 3 Not done - Not necessary |
| 188 | Checking the guardrail locking device | 1 Done - No defects |
| 189 | Checking the top limit switches and the cut-off device | 1 Done - No defects |
| 190 | Checking the upper tower crossbeam | 1 Done - No defects |
| 191 | Performing a test run | 1 Done - No defects |
| 192 | Performing the finishing steps on the service lift | 1 Done - No defects |
| | **Mechanical - Machine house** | |
| 193 | Checking the machine house anchorage points | 1 Done - No defects |
| 194 | Checking chain winch | 1 Done - No defects |
| 195 | Checking the machine house first aid kit | 1 Done - No defects |
| 196 | Checking the machine house fire extinguisher | 1 Done - No defects |
| 197 | Checking the Firespy fire-extinguishing system in the machine house | 3 Not done - Not necessary |
| 198 | Checking the machine house fitted parts | 1 Done - No defects |
| 199 | Checking the machine house casing | 1 Done - No defects |
| 200 | Checking the machine house hatches | 1 Done - No defects |
| 201 | Checking the main carrier and the axle pin | 1 Done - No defects |
| 202 | Checking yaw gears | 1 Done - No defects |
| 203 | Checking the yaw bearing | 1 Done - No defects |
| 204 | Checking the stator shield | 1 Done - No defects |
| 205 | Testing rotor lock | 1 Done - No defects |
| 206 | Checking the electromechanical brake | 1 Done - No defects |
| 207 | Checking the shaft sleeve | 1 Done - No defects |

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|---|---|---|
| 208 | Checking the machine house bolt connections | 1 Done - No defects |
| 209 | Visual inspection of machine house | 1 Done - No defects |
| | **Mechanical - Rotor head** | |
| 211 | Taking a front and rear main bearing grease sample | 1 Done - No defects |
| 212 | Greasing the rear and front rotor bearing | 1 Done - No defects |
| 213 | Checking the rotor | 2 Done - Defects detected |
| 214 | Checking the rotor head casing | 2 Done - Defects detected |
| 215 | Checking the rotor hub and the blade adapters | 2 Done - Defects detected |
| 216 | Checking rotor head seals | 1 Done - No defects |
| 217 | Checking central lubrication system on rotor head | 1 Done - No defects |
| 218 | Checking pitch gear | 1 Done - No defects |
| 219 | Checking blade flange bearing | 1 Done - No defects |
| 220 | Check lightning protection system of rotor blade A | 1 Done - No defects |
| 221 | Check lightning protection system of rotor blade B | 1 Done - No defects |
| 222 | Check lightning protection system of rotor blade C | 1 Done - No defects |
| 223 | Checking the rotor head bolt connections | 1 Done - No defects |
| 224 | Visual inspection of rotor head | 1 Done - No defects |
| | **Mechanical - Rotor blade** | |
| 225 | Checking the rotor blade A bolt connections | 1 Done - No defects |
| 226 | Checking the rotor blade B bolt connections | 2 Done - Defects detected |
| 227 | Checking the rotor blade C bolt connections | 1 Done - No defects |
| 228 | Checking the TES spring bolts on rotor blade A | 1 Done - No defects |
| 229 | Checking the TES spring bolts on rotor blade B | 1 Done - No defects |
| 230 | Checking the TES spring bolts on rotor blade C | 1 Done - No defects |
| 231 | Check rotor blade A base plate | 1 Done - No defects |
| 232 | Check rotor blade B base plate | 1 Done - No defects |
| 233 | Check rotor blade C base plate | 1 Done - No defects |
| 234 | Check laminate on rotor blade A | 1 Done - No defects |
| 235 | Check laminate on rotor blade B | 1 Done - No defects |
| 236 | Check laminate on rotor blade C | 1 Done - No defects |
| 237 | Check web bonding on rotor blade A | 1 Done - No defects |
| 238 | Check web bonding on rotor blade B | 1 Done - No defects |
| 239 | Check web bonding on rotor blade C | 1 Done - No defects |
| 240 | Check crescent on rotor blade A | 1 Done - No defects |
| 241 | Check crescent on rotor blade B | 1 Done - No defects |

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| 242 | Check crescent on rotor blade C | 1 Done - No defects |
| 243 | Check the connection between the lightning protection cable and the discharge ring on rotor blade A | 1 Done - No defects |
| 244 | Check the connection between the lightning protection cable and the discharge ring on rotor blade B | 1 Done - No defects |
| 245 | Check the connection between the lightning protection cable and the discharge ring on rotor blade C | 1 Done - No defects |
| 246 | Check lightning protection cable on rotor blade A | 1 Done - No defects |
| 247 | Check lightning protection cable on rotor blade B | 1 Done - No defects |
| 248 | Check lightning protection cable on rotor blade C | 1 Done - No defects |
| 249 | Check the connection between the lightning protection cable and the flat profile on rotor blade A | 1 Done - No defects |
| 250 | Check the connection between the lightning protection cable and the flat profile on rotor blade B | 1 Done - No defects |
| 251 | Check the connection between the lightning protection cable and the flat profile on rotor blade C | 1 Done - No defects |
| 252 | Check flat profile of web on rotor blade A | 1 Done - No defects |
| 253 | Check flat profile of web on rotor blade B | 1 Done - No defects |
| 254 | Check flat profile of web on rotor blade C | 1 Done - No defects |
| 255 | Checking the exterior of rotor blade A | 1 Done - No defects |
| 256 | Checking rotor blade B exterior | 1 Done - No defects |
| 257 | Checking rotor blade C exterior | 1 Done - No defects |
| 258 | Check flange spoiler longitudinal bolts for rotor blade A | 1 Done - No defects |
| 259 | Check flange spoiler longitudinal bolts for rotor blade B | 1 Done - No defects |
| 260 | Check flange spoiler longitudinal bolts for rotor blade C | 1 Done - No defects |
| | **Mechanical - Nacelle exterior** | |
| 261 | Checking anchorage points nacelle exterior | 1 Done - No defects |
| 262 | Checking fitted parts nacelle exterior | 2 Done - Defects detected |
| | **country-specific maintenance items** | |
| 263 | Checking rescue barrel | 1 Done - No defects |
| 264 | Check the safety ladder safety cable | 3 Not done - Not necessary |
| | **Finishing steps** | |
| 265 | Performing paintwork repairs | 1 Done - No defects |
| 266 | Acoustic check of WEC | 1 Done - No defects |
| | **Measuring device** | |
| 1000 | Type of measuring device | Fluke 1653b |
| 2000 | ID number | 1383180,0 |
| 3000 | Calibrated | on |
| 4000 | Date - month | 10,0 |

# 4-yearly maintenance

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|---|---|---|
| 5000 | Date - year | 2018,0 |
| | **Measured values for item 7** | |
| 10 | Air humidity | normal |
| 20 | Soil condition | normal |
| 30 | Outside temperature | 12,0 °C |
| 40 | Measurement 1 | 3,70 OHM |
| 50 | Measurement 2 | 3,60 OHM |
| 60 | Measurement 3 | 3,10 OHM |
| 70 | Average value | 3,50 OHM |
| 80 | Orientation of measuring probes | C |
| 90 | Test set-up with decoiler, 50m (SAP 146596) | Yes |
| | **Measured values for item 8** | |
| | **Measured values for item 10** | |
| | **Measured values for item 11** | |
| | **Measured values for item 36** | |
| 10 | Bonding bar - foundation earth electrode | 0,000 OHM |
| 20 | Bonding bar - tower entrance stairs | 0,000 OHM |
| 40 | Bonding bar - tower door frame (concrete towers only) | 0,000 OHM |
| 60 | Bonding bar - tower door socket | 0,000 OHM |
| 70 | Bonding bar - tower door industrial power socket | 0,000 OHM |
| 80 | Bonding bar - safety ladder | 0,000 OHM |
| 90 | Bonding bar - service hoist socket | 0,000 OHM |
| 110 | Bonding bar - service hoist housing | 0,000 OHM |
| 130 | Bonding bar - Tower cabling holders | 0,000 OHM |
| 180 | Bonding bar - LV distribution system =104 | 0,000 OHM |
| 200 | Bonding bar - transformer | 0,000 OHM |
| 210 | Bonding bar - grease pan of transformer | 0,000 OHM |
| 220 | Bonding bar - switchgear of transformer | 0,000 OHM |
| 230 | Bonding bar - remote control unit of MV switchgear =164 | 0,000 OHM |
| 240 | Bonding bar - Motor housing for tower fan 1 | 0,000 OHM |
| 250 | Bonding bar - Motor housing for tower fan 2 | 0,000 OHM |
| 360 | Bonding bar - control cabinet =005 | 0,000 OHM |
| 370 | Bonding bar - cabinet lighting socket in control cabinet =005 | 0,000 OHM |
| 390 | Bonding bar - UPS cabinet =068 | 0,000 OHM |
| 400 | Bonding bar - socket in UPS cabinet =069 | 0,000 OHM |

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| 410 | Bonding bar - LV-distribution system =136 terminal box | 0,000 OHM |
| 420 | Bonding bar - power cabinet =006+1 | 0,000 OHM |
| 430 | Bonding bar - power cabinet =006+2 | 0,000 OHM |
| 440 | Bonding bar - power cabinet =006+3 | 0,000 OHM |
| 450 | Bonding bar - power cabinet =006+4 | 0,000 OHM |
| 460 | Bonding bar - power cabinet =006+5 | 0,000 OHM |
| 470 | Bonding bar - power cabinet =006+6 | 0,000 OHM |
| 480 | Bonding bar - power cabinet =006+7 | 0,000 OHM |
| | **Measured values for item 37** | |
| 10 | Phase L1 - against PE of power cabinet 1 | 500,00 MOH |
| 20 | Phase L2 - against PE of power cabinet 1 | 500,00 MOH |
| 30 | Phase L3 - against PE of power cabinet 1 | 500,00 MOH |
| 40 | Phase L1 - against PE of power cabinet 2 | 500,00 MOH |
| 50 | Phase L2 - against PE of power cabinet 2 | 500,00 MOH |
| 60 | Phase L3 - against PE of power cabinet 2 | 500,00 MOH |
| 70 | Phase L1 - against PE of power cabinet 3 | 500,00 MOH |
| 80 | Phase L2 - against PE of power cabinet 3 | 500,00 MOH |
| 90 | Phase L3 - against PE of power cabinet 3 | 500,00 MOH |
| 100 | Phase L1 - against PE of power cabinet 4 | 500,00 MOH |
| 110 | Phase L2 - against PE of power cabinet 4 | 500,00 MOH |
| 120 | Phase L3 - against PE of power cabinet 4 | 500,00 MOH |
| 130 | Phase L1 - against PE of power cabinet 5 | 500,00 MOH |
| 140 | Phase L2 - against PE of power cabinet 5 | 500,00 MOH |
| 150 | Phase L3 - against PE of power cabinet 5 | 500,00 MOH |
| 160 | Phase L1 - against PE of power cabinet 6 | 500,00 MOH |
| 170 | Phase L2 - against PE of power cabinet 6 | 500,00 MOH |
| 180 | Phase L3 - against PE of power cabinet 6 | 500,00 MOH |
| 190 | Phase L1 - against PE of power cabinet 7 | 500,00 MOH |
| 200 | Phase L2 - against PE of power cabinet 7 | 500,00 MOH |
| 210 | Phase L3 - against PE of power cabinet 7 | 500,00 MOH |
| | **Measured values for item 41** | |
| | **Measured values for item 42** | |
| | **Measured values for item 51** | |
| 10 | Socket next to entrance door, phase angle 0° | 36,00 MSC |
| 20 | Socket next to entrance door, phase angle 0° | 0,0 V |

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| 30 | Socket next to entrance door, phase angle 180° | 27,00 MSC |
| 40 | Socket next to entrance door, phase angle 180° | 0,0 V |
| 50 | Socket next to entrance door, phase angle 0° | 24,00 MA |
| 60 | Socket next to entrance door, phase angle 0° | 0,0 V |
| 70 | Socket next to entrance door, phase angle 180° | 24,00 MA |
| 80 | Socket next to entrance door, phase angle 180° | 0,0 V |
| | **Measured values for item 52** | |
| | **Measured values for item 53** | |
| 10 | Power cabinet 1 output L1 | 7500,0 A |
| 20 | Power cabinet 1 output L2 | 7500,0 A |
| 30 | Power cabinet 1 output L3 | 7500,0 A |
| 40 | Control cabinet =005 output line contactor L1 | 7500,0 A |
| 50 | Control cabinet =005 output line contactor L2 | 7500,0 A |
| 60 | Control cabinet =005 output line contactor L3 | 7500,0 A |
| | **Measured values for item 57** | |
| 10 | Service hoist socket, phase angle 0° | 39,80 MSC |
| 20 | Service hoist socket, phase angle 0° | 0,0 V |
| 30 | Service hoist socket, phase angle 180° | 47,20 MSC |
| 40 | Service hoist socket, phase angle 180° | 0,0 V |
| 50 | Service hoist socket, phase angle 0° | 24,00 MA |
| 60 | Service hoist socket, phase angle 0° | 1,0 V |
| 70 | Service hoist socket, phase angle 180° | 30,00 MA |
| 80 | Service hoist socket, phase angle 180° | 0,0 V |
| | **Measured values for item 59** | |
| 10 | Phase L1 industrial power socket, tower door - PE | 500,00 MOH |
| 20 | Phase L2 industrial power socket, tower door - PE | 500,00 MOH |
| 30 | Phase L3 industrial power socket, tower door - PE | 500,00 MOH |
| 40 | Neutral conductor industrial power socket, tower door - PE | 200,00 MOH |
| 50 | Phase L1 against phase L2 industrial power socket, tower door - PE | 500,00 MOH |
| 60 | Phase L1 against phase L3 industrial power socket, tower door - PE | 500,00 MOH |
| 70 | Phase L2 against phase L3 industrial power socket, tower door - PE | 500,00 MOH |
| 80 | Phase of socket, tower door - PE | 200,00 MOH |
| 90 | Neutral conductor of socket, tower door - PE | 200,00 MOH |
| 120 | Phase of socket, in service hoist - PE | 200,00 MOH |
| 130 | Neutral conductor of socket in service hoist - PE | 200,00 MOH |

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| 200 | Phase of cabinet lighting socket in control cabinet =005 - PE | 200,00 MOH |
| 210 | Neutral conductor of cabinet lighting socket in control cabinet - PE | 200,00 MOH |
| 220 | Socket phase in UPS cabinet =068 - PE | 200,00 MOH |
| 230 | Socket, neutral wire in UPS cabinet =068 - PE | 200,00 MOH |
| | **Measured values for item 61** | |
| 10 | Phase of upper socket in tower - PE | 200,00 MOH |
| 20 | Neutral conductor of upper socket in tower - PE | 2,20 MOH |
| 30 | Phase L1 of upper industrial power socket in tower - PE | 200,00 MOH |
| 40 | Phase L2 of upper industrial power socket in tower - PE | 200,00 MOH |
| 50 | Phase L3 of upper industrial power socket in tower - PE | 200,00 MOH |
| 60 | Neutral conductor of upper industrial power socket in tower - PE | 2,20 MOH |
| 70 | Phase L1 against phase L2 of upper industrial power socket in tower | 500,00 MOH |
| 80 | Phase L1 against phase L3 of upper industrial power socket in tower | 500,00 MOH |
| 90 | Phase L2 against phase L3 of upper industrial power socket in tower | 500,00 MOH |
| | **Measured values for item 62** | |
| 10 | Topmost socket in tower, 0° | 36,00 MSC |
| 20 | Topmost socket in tower, 0° | 0,0 V |
| 30 | Topmost socket in tower, 180° | 26,90 MSC |
| 40 | Topmost socket in tower, 180° | 0,0 V |
| 50 | Top socket in tower, 0° | 24,00 MA |
| 60 | Top socket in tower, 0° | 0,0 V |
| 70 | Top socket in tower, 180° | 27,00 MA |
| 80 | Top socket in tower, 180° | 0,0 V |
| | **Measured values for item 71** | |
| 10 | Phase L1 of industrial power socket 1 on nacelle control cabinet - PE | 200,00 MOH |
| 20 | Phase L2 of industrial power socket 1 on nacelle control cabinet - PE | 200,00 MOH |
| 30 | Phase L3 of industrial power socket 1 on nacelle control cabinet - PE | 200,00 MOH |
| 40 | Neutral conductor of industrial power socket 1 nacelle control cabinet - PE | 200,00 MOH |
| 50 | Phase L1 against L2 of industrial power socket 1 on nacelle control cabinet | 500,00 MOH |
| 60 | Phase L1 against L3 of industrial power socket 1 on nacelle control cabinet | 500,00 MOH |
| 70 | Phase L2 against L3 of industrial power socket 1 on nacelle control | 500,00 MOH |
| 100 | Phase of socket on nacelle control cabinet - PE | 200,00 MOH |
| 110 | Neutral conductor of socket on nacelle control cabinet - PE | 200,00 MOH |
| 120 | Phase of socket in main carrier - PE | 200,00 MOH |
| 130 | Neutral conductor of socket in main carrier - PE | 200,00 MOH |

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| | **Measured values for item 73** | |
| 10 | Socket on nacelle control cabinet, phase angle 0° | 20,30 MSC |
| 20 | Socket on nacelle control cabinet, phase angle 0° | 1,0 V |
| 30 | Socket on nacelle control cabinet, phase angle 180° | 27,10 MSC |
| 40 | Socket on nacelle control cabinet, phase angle 180° | 0,0 V |
| 50 | Socket on nacelle control cabinet, phase angle 0° | 24,10 MA |
| 60 | Socket on nacelle control cabinet, phase angle 0° | 0,0 V |
| 70 | Socket on nacelle control cabinet, phase angle 180° | 24,00 MA |
| 80 | Socket on nacelle control cabinet, phase angle 180° | 1,0 V |
| | **Measured values for item 91** | |
| 10 | Nacelle control cabinet, input nacelle power supply L1 | 3800,0 A |
| 20 | Nacelle control cabinet, input nacelle power supply L2 | 3800,0 A |
| 30 | Nacelle control cabinet, input nacelle power supply L3 | 2500,0 A |
| | **Measured values for item 92** | |
| 20 | Bonding bar - industrial power socket on main carrier | 0,000 OHM |
| 30 | Bonding bar - socket in main carrier | 0,000 OHM |
| 50 | Bonding bar - excitation controller box =018 | 0,000 OHM |
| 60 | Bonding bar - fan, frequency converter cabinet =088+1 | 0,000 OHM |
| 70 | Bonding bar - Yaw inverter cabinet = 102 | 0,000 OHM |
| 80 | Bonding bar - nacelle control cabinet =015 | 0,000 OHM |
| 90 | Bonding bar - industrial power socket 1 on nacelle control cabinet | 0,000 OHM |
| 100 | Bonding bar - socket on nacelle control cabinet | 0,000 OHM |
| 130 | Bonding bar - winch motor housing | 0,000 OHM |
| 140 | Bonding bar - beacon system control cabinet=115 | 0,000 OHM |
| 150 | Bonding bar - yaw motor 1 | 0,000 OHM |
| 160 | Bonding bar - yaw motor 2 | 0,000 OHM |
| 170 | Bonding bar - yaw motor 3 | 0,000 OHM |
| 180 | Bonding bar - yaw motor 4 | 0,000 OHM |
| 190 | Bonding bar - yaw motor 5 | 0,000 OHM |
| 200 | Bonding bar - yaw motor 6 | 0,000 OHM |
| 210 | Bonding bar - rectifier cabinet =017+1 | 0,000 OHM |
| 220 | Bonding bar - rectifier cabinet =017+2 | 0,000 OHM |
| 230 | Bonding bar - filter cabinet =066 part 1.1 | 0,000 OHM |
| 240 | Bonding bar - filter cabinet =066 part 1.2 | 0,000 OHM |
| 250 | Bonding bar - filter cabinet =066 part 2.1 | 0,000 OHM |

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| 260 | Bonding bar - filter cabinet =066 part 2.2 | 0,000 OHM |
| 270 | Bonding bar - Machine house casing (only for aluminium nacelle casing) | 0,000 OHM |
| 280 | Bonding bar - stator sub distribution system =014 | 0,000 OHM |
| 290 | Bonding bar - stator | 0,000 OHM |
| 300 | Bonding bar - rotor (brake disk) | 0,000 OHM |
| 310 | Bonding bar - rotor brake =147+A | 0,000 OHM |
| 320 | Bonding bar - rotor brake =147+B | 0,000 OHM |
| 330 | Bonding bar - motor housing for generator air 1 | 0,000 OHM |
| 340 | Bonding bar - motor housing for generator air 2 | 0,000 OHM |
| 350 | Bonding bar - motor housing for generator air 3 | 0,000 OHM |
| 360 | Bonding bar - motor housing for generator air 4 | 0,000 OHM |
| 370 | Bonding bar - motor housing for generator air 5 | 0,000 OHM |
| 380 | Bonding bar - motor housing for generator air 6 | 0,000 OHM |
| 390 | Bonding bar - surge protection device, generator =019 | 0,000 OHM |
| 400 | Bonding bar - residual current monitoring, generator =148 | 0,000 OHM |
| 410 | Bonding bar - anemometer suppor frame 1 | 0,000 OHM |
| | **Measured values for item 94** | |
| 10 | Air humidity | 42,00 % |
| 20 | Stator temperature | 40,00 °C |
| 30 | Rotor temperature | 39,00 °C |
| 40 | Unit =017 single phase from System 1 to PE (1min) | 1800,000 MOH |
| 50 | Unit =017 single phase from System 1 to PE (10min) | 9500,000 MOH |
| 60 | Unit =017 single phase from System 1 to PE [PI] | 5,260 |
| 70 | Unit =017 single phase from System 2 to PE (1min) | 1230,000 MOH |
| 80 | Unit =017 single phase from System 2 to PE (10min) | 5600,000 MOH |
| 90 | Unit =017 single phase from System 2 to PE [PI] | 4,600 |
| 100 | Unit =017 single phase from System 1 to System 2 (1min) | 2600,000 MOH |
| 110 | Unit =017 single phase from System 1 to System 2 (10min) | 12500,000 MOH |
| 120 | Unit =017 single phase from System 1 to System 2 [PI] | 4,760 |
| | **Measured values for item 98** | |
| 10 | Rotor sub-distribution system =013 L1 -FC04:1 | 229,0 A |
| | **Measured values for item 99** | |
| 10 | Bonding bar - spinner lighting | 0,000 OHM |
| 20 | Bonding bar - blade relay box A =011 | 0,000 OHM |
| 30 | bonding bar - pitch control box A =012 | 0,000 OHM |

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| 40 | Bonding bar - compact limit switch A =028 | 0,000 OHM |
| 60 | Bonding bar - pitch motor 1 A =053 | 0,000 OHM |
| 70 | Bonding bar - load control box =075+A (LoadControl) | 0,000 OHM |
| 80 | Bonding bar - rotor blade A capacitor box =076 | 0,000 OHM |
| 90 | Bonding bar - blade relay box B =011 | 0,000 OHM |
| 100 | Bonding bar - pitch control box B =012 | 0,000 OHM |
| 110 | Bonding bar - compact limit switch B =028 | 0,000 OHM |
| 120 | Bonding bar - pitch motor 1 B =053 | 0,000 OHM |
| 130 | Bonding bar - load control box =075+B (LoadControl) | 0,000 OHM |
| 140 | Bonding bar - rotor blade B capacitor box =076 | 0,000 OHM |
| 150 | Bonding bar - blade relay box C =011 | 0,000 OHM |
| 160 | Bonding bar - pitch control box C =012 | 0,000 OHM |
| 170 | Bonding bar - compact limit switch C =028 | 0,000 OHM |
| 180 | Bonding bar - pitch motor 1 C =053 | 0,000 OHM |
| 190 | Bonding bar - load control box =075+C (LoadControl) | 0,000 OHM |
| 200 | Bonding bar - rotor blade C capacitor box =076 | 0,000 OHM |
| 210 | Bonding bar - rotor sub-distribution system =013 | 0,000 OHM |
| 220 | Bonding bar - socket on the rotor | 0,000 OHM |
| 230 | Bonding bar - hub sub-distribution system =035 | 0,000 OHM |
| 240 | Bonding bar - slip ring unit =051 | 0,000 OHM |
| 250 | Bonding bar - transformer cabinet =133 | 0,000 OHM |
| 260 | Bonding bar - hub | 0,000 OHM |
| 270 | Bonding bar - rotor head casing (with PE cable connected) | 0,000 OHM |
| 280 | Bonding bar - earthing cable - rotor head | 0,000 OHM |
| | **Measured values for item 101** | |
| 10 | Phase of socket, on the rotor - PE | 200,00 MOH |
| 20 | Neutral conductor of socket on the rotor - PE | 200,00 MOH |
| | **Measured values for item 102** | |
| 10 | Socket on the rotor, phase angle 0° | 16,30 MSC |
| 20 | Socket on the rotor, phase angle 0° | 1,0 V |
| 30 | Socket on the rotor, phase angle 180° | 24,60 MSC |
| 40 | Socket on the rotor, phase angle 180° | 0,0 V |
| 50 | Socket on the rotor, phase angle 0° | 15,00 MA |
| 60 | Socket on the rotor, phase angle 0° | 0,0 V |
| 70 | Socket on the rotor, phase angle 180° | 15,00 MA |

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| 80 | Socket on the rotor, phase angle 180° | 1,0 V |
| | **Measured values for item 134** | |
| | **Measured values for item 160** | |
| 10 | next (third-party) inspection - month | 1,0 |
| 20 | next (third-party) inspection - year | 2020,0 |
| | **Measured values for item 180** | |
| 10 | Winch | Goracon G-Trac 600 (60917) |
| 20 | Operating hours | 2,0 H |
| 30 | YOM of winch | 2018,0 |
| | **Measured values for item 194** | |
| 10 | YOM of winch | 2010,0 |

## Findings / Notification overview

| No. | Defect location | Defect type | Notification |
|---|---|---|---|
| 2 | special entries - defect location not found | special entries - unknown, fault localisation | 920036858365 |
| 60 | tower, inside fitted part - socket, platform 230V | special entries - defect type not found | 920036858364 |
| 83 | nacelle, machine house - cabinet, generator filter inside | special, electrical component - fuse defect | 920036473536 |
| 89 | nacelle, machine house - cabinet, nacelle control | special entries - defect type not found | 920036853669 |
| 116 | rotor system - discharge box | connection, earthing - PE connection not fitting, too short | 920036482140 |
| 150 | tower, safety ladder, concrete tower - rung | fit/gap size - distance/gap size too small | 920036858085 |
| 213 | generator, rotor, fitted part - access tarpaulin | special entries - defect type not found | 920036858086 |
| 214 | rotor system - front, casing | dirt/waste - foreign object | 920029647312 |
| 215 | special entries - defect location not found | special entries - defect type not found | 920036858087 |
| 226 | Base plate - Bracket Cross bolt - base plate | Coating damage - Crack, lateral | 920036858363 |
| 262 | outside, nacelle, fitted part - anemometer, support, carrier tube | connection, screwing - wrong | 920036483975 |

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